Chinnadandluru Vishnu Vardhan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.18
BHEERAM SREEDHR REDDY INTERNATIONAL SCHOOL	Class XII (CBSE)	2022	97.60%
BHEERAM SREEDHR REDDY INTERNATIONAL SCHOOL	Class X (CBSE)	2024	94.20%

PROJECTS

6-DOF Robotic Arm Simulation | MATLAB,Simulink, SolidWork

- Developed a **6-DOF robotic arm simulation in MATLAB** by importing and integrating CAD models designed in **SolidWorks**.
- Modeled the robot's **kinematic structure and joint motion** to simulate the movement of individual links and the overall manipulator.
- Analyzed **forward kinematics and joint-space motion** to visualize the robotic arm configuration and end-effector movement.
- Integrated CAD geometry with the MATLAB simulation environment to achieve a **realistic 3D representation of the robotic system**.

Wireless ESP32-Based RC Car

- contributed to an **ESP32-based RC car** with smartphone-controlled wireless operation for forward, reverse, left, and right motion.
- **Integrated the ESP32, L298N motor driver, DC motors, and rechargeable 2S Li-ion battery pack** into a compact vehicle control and power architecture.
- **Configured the motor-control and power interfaces** to achieve reliable wireless command execution and bidirectional vehicle movement.

Six-Bar Quick Return Mechanism | Kinematics & Dynamics of Machinery

- Designed and developed a **six-bar quick-return mechanism** to achieve unequal forward and return strokes.
- Performed **kinematic and dynamic analysis** to evaluate link positions, velocities, accelerations, and forces throughout the motion cycle.

3D CAD Modeling & Additive Manufacturing

- Designed a **3D model using CAD software**, focusing on dimensional accuracy and printability.
- Fabricated the designed model using **FDM 3D printing**, optimizing print orientation and support requirements.

POSITIONS OF RESPONSIBILITY

Head, Torque-Automobile club, IIT Hyderabad

- Coordinate club operations, responsibilities to core members, and mentor cores and junior students in technical activities
- Initiated new learning sessions and project-based activities to increase student engagement in automotive engineering.

Core Member, Torque -Automobile Club, IIT Hyderabad

- Collaborated with fellow core members to improve event planning, publicity, and smooth execution of club initiatives.
- Assisted in organizing technical workshops, and outreach activities while coordinating logistics, and event execution.

Hospitality Coordinator, Milan-Inter Hostel Fest, IIT Hyderabad

- Worked closely with multiple organizing committees to resolve logistical challenges and ensure seamless event execution within strict timelines.
- Coordinated volunteer teams, managed task allocation, and monitored event operations to ensure smooth execution of hospitality activities.

SKILLS

- Simulation and Analysis Tools : MATLAB, Ansys Mechanical ,Abaqus
- CAD & Design : Solid edge, SolidWorks
- Programming Languages : Python, C
- Basics of Machine Learning (Supervised Learning): Regression, Classification, Basic ML Algorithms
- Libraries : Numpy, Matplotlib
- Microcontrollers : Arduino

RELEVANT COURSES

- **Computer Science** : Introduction to Programming
- **Mathematics** : Elementary Linear Algebra , Introduction to Statistics, Differential Equations, Introduction to Probability
- **Mechanical Core courses :** Engineering Drawing, Engineering mechanics, Solid mechanics, Fluid mechanics, Heat and Mass Transfer, Kinematics and Dynamics in Machinery, Manufacturing science

EXTRACURRICULAR

NSO Hockey

- Participated in the institute's Hockey League (IHL) as an NSO Hockey member.